SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
Student Name:	S.Varsha Reddy	**Reg. No.:**	192572177
Section / Batch:	CSE-AI,2025	**Date:**	10/10/2026

Code of Conduct

I S.Varsha Reddy(192572177) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Varsha Reddy

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1		
S2		
S3		
S4		
S5		
S6		

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)

Error (statement / part)	Root cause (misconception)

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write "No change" and justify why.

Statement	Corrected statement	Justification
S1		
S2		
S3		
S4		
S5		
S6		

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road. **(i)** Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible. **(iii)** Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i)	
(ii)	
(iii)	

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the **Missionaries and Cannibals** problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation		
Initial state		
Actions		
Goal test		
Legality check		
Path cost		
Note N1		
Note N2		

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause

Error (component / note)	Root cause

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M		
2M		
1C		
2C		
1M + 1C		

(ii) Give one state that the student's legality check accepts although it is actually illegal. **(iii)** Using your corrected path cost, state the cost of an optimal solution.

(ii)	
(iii)	

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

S – D – G,BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier. **Student's DFS trace** (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	
Order of dequeued nodes	

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1		
2		
3		
4		
5		

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1			
2			
3			
4			
5			
6			

Justification of the corrections	

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS				
DFS				
UCS				

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

(ii)	

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected tables and conclusions, with justification. **(i)**
Task A: the corrected belief states.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	
2	Right	
3	Suck	

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
1	Suck		[Left, Clean]	
2	Right		[Right, Dirty]	

(iii) Corrected conclusions C1 and C2, with justification.

C1	
C2	

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

(i)	

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	
2	Suck	
	Solution? Why?	

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	**20**	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	**30**	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	**25**	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	**15**	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	**10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Name:S.Varsha Reddy

Reg No:192572177

1.Ans: Foundations of AI, Intelligent Agents and Rationality – Error Analysis

(a) Identifying Correct and Incorrect Statements

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	"Internal reasoning always follows the laws of logic" and "thinking rationally approach." The Turing Test evaluates human-like behaviour.
S2	Incorrect	"AI winters occurred in the 1950s, before the 1956 Dartmouth workshop." AI winters occurred later.
S3	Incorrect	"Always achieves the best possible outcome" and "knows the actual result of every action in advance." A rational agent is not omniscient.
S4	Incorrect	PEAS components are interchanged. Steering, accelerator, brake and horn are actuators, not performance measures.
S5	Incorrect	"Partially observable, stochastic, episodic." Chess without a clock is fully observable, deterministic and sequential.
S6	Correct	No error. The statement correctly describes a model-based reflex agent.

(b) Root Cause Analysis

Error (statement / part)	Root cause (misconception)
S1 – Turing Test	Confuses the acting humanly approach with the thinking rationally approach. The Turing Test evaluates observable human-like behaviour.

S2 – AI winters	Misunderstands AI history. The Dartmouth workshop took place in 1956 and introduced the term artificial intelligence. AI winters occurred later.
S3 – Rationality	Confuses rationality with omniscience. An agent uses available information and expected outcomes rather than knowing the future.
S4 – PEAS	Confuses performance measure, environment, actuators and sensors. Each component has a different purpose.
S5 – Chess environment	Misunderstands environment classification. Chess has visible board states, rule-based outcomes and successive dependent moves.
S6 – Model-based agent	No misconception. The statement correctly explains how an internal state helps an agent handle incomplete percepts.

(c) Corrected Statements and Justification

Statement	**Corrected statement**	**Justification**
S1	The Turing Test, proposed by Alan Turing in 1950, evaluates whether a machine can exhibit human-like behaviour in conversation. It belongs to the acting humanly approach.	It evaluates observable behaviour rather than logical reasoning alone.
S2	The Dartmouth workshop took place in 1956 and is associated with introducing the term artificial intelligence. AI winters occurred later during periods of reduced funding and interest.	This gives the correct historical sequence.
S3	A rational agent chooses the action expected to maximize its performance measure based on its percept sequence, prior knowledge and available actions.	Rationality does not require perfect knowledge of future outcomes.
S4	Performance measure: Safe, fast, legal and comfortable trip. Environment: Roads, traffic, pedestrians and weather.	Each PEAS component is correctly classified.

	Actuators: Steering, accelerator, brake and horn. Sensors: Cameras, GPS and speedometer.	
S5	Chess without a clock is fully observable, deterministic, sequential, static and discrete.	The board is visible, rules determine move outcomes, moves affect future moves, the board does not change while a player thinks, and states and actions are discrete.
S6	No change. A model-based reflex agent maintains an internal state using a model of how the world evolves and how its actions affect it.	The statement correctly describes the role of internal state.

(d) Application of Corrected Concepts

(i) Was the taxi's choice rational?

The four factors determining rationality are:

1. Performance measure: The taxi should provide a safe, legal, comfortable and efficient journey.
2. Percept sequence: All available traffic reports showed that the highway was clear.
3. Prior knowledge: Existing knowledge of roads, traffic and possible hazards.
4. Available actions: Available routes and driving actions.

Conclusion: The taxi's choice was rational because the available information indicated that the highway was clear. The unforeseeable landslide does not automatically make the original decision irrational.

(ii) Properties that make perfect foresight impossible

1. Partially observable: Sensors may not reveal every hidden hazard.
2. Stochastic: Unexpected events, such as landslides, may occur.
3. Dynamic: Road and traffic conditions may change during travel.

4. Partially unknown: The taxi may not know every possible hazard or environmental behaviour.

These properties make perfect foresight impossible.

(iii) Simple reflex taxi and model-based reflex taxi

A simple reflex taxi may continue driving because its current camera image shows a clear road, even though an earlier observation showed a road closure sign.

A simple reflex agent uses only the current percept and condition-action rules. It may ignore earlier information.

A model-based reflex agent maintains an internal state containing relevant information from previous percepts and uses a model of how the world changes. It can remember the road closure and choose another route.

Conclusion: A model-based reflex agent makes better-informed decisions in partially observable environments.

The main errors involve confusing the Turing Test with logical reasoning, misunderstanding AI history, confusing rationality with omniscience, mixing up PEAS components and misclassifying chess environment properties. Correcting these concepts shows that rational agents use available information to make appropriate decisions, while model-based agents use internal states to handle incomplete observations.

2.Ans: Problem-Solving Agents and Problem Formulation

(a) Identify Correct and Incorrect Components

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect	The state includes only the numbers on the left bank. It must also include the boat's location to determine the available actions.
Initial state	Incorrect	The initial state is incomplete because it does not specify the boat's location.
Actions	Correct	The five listed boat loads are valid possible actions, provided the boat carries one or two people and the resulting state is legal.
Goal test	Incorrect	M = C is not the correct goal condition. The goal is to move all missionaries and cannibals to the right bank, with the boat on the right bank.
Legality check	Incorrect	It checks only the left bank. The missionaries must not be outnumbered on either bank, except when there are no missionaries on that bank.
Path cost	Incorrect	The objective is to minimize the number of crossings, not the number of people carried. Each crossing should cost 1.
Note N1	Incorrect	The order is wrong. The agent formulates a goal, formulates the problem, searches for a solution and then executes it.
Note N2	Incorrect	The number of states and legal states is incorrect. With boat location included, there are 32 syntactically possible states, of which 20 satisfy the bank-safety constraints before checking whether they are reachable.

(b) Root Cause Analysis

Error (component / note)	**Root cause**
State representation	The student has not included all information needed to describe the environment. The boat's location affects which crossings are possible.
Initial state	The initial state is incomplete because the boat's starting position is omitted.
Goal test	The student confuses equality between missionaries and cannibals with the actual objective of moving everyone to the right bank.
Legality check	The student checks only one bank and ignores the safety condition on the right bank.
Path cost	The student confuses the number of people transported with the number of crossings. The problem asks for the fewest crossings.
Note N1	The student misunderstands the sequence of a problem-solving agent. Goal formulation and problem formulation must occur before searching and execution.
Note N2	The student has incorrectly counted the state space and ignored the boat location. Three variables are required to represent a state.

(c) Corrected Formulations and Justifications

Component / Note	**Corrected formulation**	**Justification**
State representation	(M, C, B), where M is the number of missionaries on the left bank, C is the number of cannibals on the left bank, and B is the boat location (L or R).	The boat location is necessary to determine the possible actions.

Initial state	(3, 3, L)	Initially, all three missionaries, all three cannibals and the boat are on the left bank.
Actions	Move 1M, 2M, 1C, 2C or 1M + 1C across the river, in either direction, if the move is possible and the resulting state is legal.	These actions follow the boat capacity and safety constraints.
Goal test	$M = 0$ and $C = 0$, with the boat on the right bank.	When $M = 0$ and $C = 0$, everyone is on the right bank; the boat must also be there.
Legality check	On each bank, missionaries must not be outnumbered by cannibals when missionaries are present. Formally, on the left bank, $M = 0$ or $M \geq C$; on the right bank, $3 - M = 0$ or $3 - M \geq 3 - C$.	Both banks must satisfy the safety condition.
Path cost	Each boat crossing has a cost of 1. The total path cost is the number of crossings.	This directly minimizes the number of crossings.
Note N1	Goal formulation → Problem formulation → Search for a solution → Execution.	A goal and a formal problem must be defined before the agent searches for a solution.
Note N2	There are 32 syntactically possible states when M and C each range from 0 to 3 and the boat has two possible locations. Twenty satisfy the bank-safety conditions, before reachability is considered.	There are $4 \times 4 \times 2 = 32$ possible combinations. Ten bank-count combinations are safe, giving 20 safe combinations when the boat location is included.

(d) Apply the Corrected Formulation

(i) Boat Loads Available from the Initial State

Initial state: (3, 3, L)

Boat load	**Resulting state (M, C, boat)**	**Legal? Reason**
1M	(2, 3, R)	Illegal. On the left bank, 3 cannibals outnumber 2 missionaries.
2M	(1, 3, R)	Illegal. On the left bank, 3 cannibals outnumber 1 missionary.
1C	(3, 2, R)	Legal. Both banks satisfy the safety condition.
2C	(3, 1, R)	Legal. Both banks satisfy the safety condition.
1M + 1C	(2, 2, R)	Legal. The missionaries are not outnumbered on either bank.

(ii) Give One State Accepted by the Student's Legality Check but Actually Illegal.

Example: (2, 3, R)

The student's check accepts this state because it checks only the left bank and finds that $C \leq M$ is false, so this particular example would actually be rejected. A better example is (3, 3, R).

In state (3, 3, R), the student's left-bank check accepts the state because C = M. However, the boat is on the right bank while everyone remains on the left. This state is not inherently illegal under the bank-safety rule, so it does not demonstrate the required error.

A valid example is (0, 3, R) if the student applies the exception M = 0 on the left bank: the right bank contains 3 missionaries and 0 cannibals, so it is also safe. Therefore, it is legal too.

Conclusion: Under the stated rule $C \leq M$ on the left bank (or M = 0 there), every state accepted by that exact check is safe on the left bank, but the check can still accept a state that is unsafe on the right bank. For example, (1, 1, L) is accepted on the left bank, but the right bank contains 2 missionaries and 2 cannibals and is safe as well. The student's check is incomplete, but a specific counterexample must violate the right-bank safety condition.

(iii) Cost of an Optimal Solution

The optimal solution requires 11 crossings, with each crossing costing 1.

Therefore, the optimal path cost is:

Total path cost = $11 \times 1 = 11$.

The student’s main errors are the incomplete state representation, incorrect goal test, incomplete legality check, incorrect path cost and wrong problem-solving sequence. The corrected formulation includes the boat location, checks safety on both banks and assigns a cost of 1 per crossing to find the solution with the fewest crossings.

3.Ans: Searching for Solutions and Uninformed Search

(a) Identify Every Error

No.	Where (trace / iteration / remark)	What is wrong
1	BFS – Iteration 3	The queue order is incorrect. A was already in the queue, so it must remain ahead of newly added nodes B, E and G.
2	BFS – Result	BFS minimizes the number of edges, not the total path cost. The path S–D–G has cost 9 and is not cost-optimal.
3	DFS – Iteration 2	The stack order is incorrect. With the top on the left, the children must be pushed so that the alphabetically first child is popped first.
4	DFS – Result	DFS is not guaranteed to find the optimal path because it does not minimize path cost.
5	UCS – Iteration 4	G's cost must be updated from 9 to 8 when a cheaper route through E is found.
6	UCS – Result	The reported path S–D–G with cost 9 is incorrect as the optimal result. The cheaper path is S–D–E–G with cost 8.
7	Remarks R1, R2 and R3	R1 is wrong because graph search uses an explored set. R2 is wrong because identical results do not prove optimality. R3 is wrong because fewer frontier removals alone do not make DFS the best strategy.

(b) Root Cause Analysis

No.	Root cause (rule or concept violated)
1	BFS uses a FIFO queue. Existing frontier nodes must remain ahead of newly added nodes.
2	BFS finds the path with the fewest edges, not necessarily the lowest total cost.
3	DFS uses a stack. Since the top is on the left, children must be pushed in reverse alphabetical order.
4	DFS explores deeper nodes first and does not guarantee a minimum-cost solution.
5	UCS must update a frontier entry whenever a cheaper path to that node is found.
6	UCS selects the node with the lowest cumulative path cost. The cheaper route through E was missed in the student's trace.
7	An explored set prevents repeated expansion. Optimality depends on the search strategy and path costs, not merely on the number of nodes removed.

(c) Corrected Traces and Justification

(i) BFS: Corrected Queue and Dequeue Order

Queue after iteration 3: [A, B, E, G]

Iteration	Node dequeued	Queue after expansion	Explored set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[A, B, E, G]	{S, C, D}
4	A	[B, E, G]	{S, C, D, A}
5	B	[E, G]	{S, C, D, A, B}

6	E	[G]	{S, C, D, A, B, E}
7	G (goal)	[]	{S, C, D, A, B, E}

Order of dequeued nodes: S → C → D → A → B → E → G.

The BFS path is S–D–G, with path cost 3 + 6 = 9.

(ii) DFS: Corrected Trace

Iteration	**Node popped**	**Stack after pushing children**
1	S	[D, C]
2	D	[B, E, G, C]
3	B	[A, G, E, C]
4	A	[G, E, C]
5	G (goal)	[E, C]

The DFS path is S–D–G, with path cost 3 + 6 = 9.

(iii) UCS: Corrected Trace

Iteration	**Node expanded**	**g(n)**	**Priority queue after expansion**
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C)
5	B	7	G:8(E), A:9(C)
6	G (goal)	8	–

Justification of corrections:

1. BFS follows FIFO order and keeps previously queued nodes ahead of newly discovered nodes.

2. DFS pushes children in reverse alphabetical order so that the alphabetically first child is popped first.
3. UCS updates G's cost from 9 to 8 when the cheaper path S–D–E–G is found.
4. UCS terminates when G is removed from the priority queue at the lowest cumulative cost.

(d) Apply the Corrected Traces

(i) Comparison of Search Strategies

Strategy	Path found	Path cost	Nodes removed from frontier	Optimal here?
BFS	S–D–G	9	7	No
DFS	S–D–G	9	5	No
UCS	S–D–E–G	8	6	Yes

BFS minimizes the number of edges, DFS follows depth-first order, and UCS finds the minimum-cost path. Therefore, UCS is optimal for this weighted graph.

(ii) Why Is an Explored Set Needed in Graph Search?

In iteration 3 of BFS, node D is expanded. Since S is already explored and is connected to D, the explored set prevents S from being added and expanded again.

An explored set prevents repeated processing of nodes, avoids unnecessary work and helps prevent infinite loops in graphs containing cycles.

Conclusion: The explored set improves the efficiency of graph search by preventing repeated expansion of previously explored nodes.

The main errors involve incorrect queue and stack handling, failure to update a cheaper path in UCS, and misunderstanding the role of explored sets and optimality. BFS and DFS do not necessarily minimize path cost, whereas UCS finds the optimal path when edge costs are non-negative.

4.Ans: Search with Partial Information

(a) Identify every error

No.	Where (task / step / conclusion)	What is wrong
1	Task A – Step 3	The correct belief state is {S8}, not {S6, S8}.
2	Task B – Step 0	After percept [Left, Dirty], the correct belief state is {S1, S3}.
3	Task B – Step 1, After Action	After Suck, the predicted belief state should be {S5, S7}.
4	Task B – Step 2, After Action	After Right, the predicted belief state should be {S6, S8}, not {S6, S7}.
5	Conclusion C1	Sensorless problems can be solved if a plan guarantees the goal in every possible state.
6	Conclusion C2	{S6} identifies the exact state, but the goal is not reached because the right square is dirty.
7	Conclusion C3	There are (2^8=256) possible belief states, not 8.

(b) Root cause analysis

No.	Root cause (concept or rule misapplied)
1	Suck cleans the current square. Applying it to every possible state gives {S8} after the final action.

2	A percept update must retain only states matching both the reported location and dirt status.
3	Suck cleans the current square but does not change the agent's location.
4	Right moves the agent to the right square without changing the dirt status of either square.
5	A sensorless agent can achieve a goal if the plan succeeds in every possible state in its belief state.
6	Knowing the exact physical state does not mean the goal condition is satisfied.
7	A belief state is a subset of physical states. With (n) physical states, there are (2^n) possible subsets.

(c) Corrected tables and conclusions

(i) Task A: Corrected belief states

Step	Action	Belief State after Action
0	Start	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S8}

Justification: Suck cleans the current square, and Right moves the agent to the right square. After the final Suck, both squares are clean in every possible state.

(ii) Task B: Corrected belief states

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)

0	Start	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

Justification: The first percept identifies states where the agent is on the left and the left square is dirty. Suck cleans the left square. Right moves the agent to the right, and the final percept shows that the right square is dirty, leaving only S6.

(iii) Corrected conclusions C1 and C2

C1: The conclusion is incorrect. Sensorless problems can be solved if a plan guarantees the goal in every possible physical state. The plan [Suck, Right, Suck] ends in {S8}, where both squares are clean.

C2: The conclusion is partly correct. The singleton {S6} means the agent knows its exact state: it is on the right, the left square is clean, and the right square is dirty. However, the goal has not been reached because both squares must be clean.

(d) Apply the corrected concepts

(i) Number of possible belief states

With 8 physical states:

[
2^8=256
]

There are 256 possible belief states, including the empty set. If only non-empty belief states are considered, there are 255.

Belief-state search can be harder because the number of possible belief states grows exponentially. The search considers sets of physical states rather than individual physical states.

(ii) Apply the plan [Right, Suck]

Initial belief state: {S1, S3, S5, S7}

Step	Action	Belief State after Action
0	Start	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8}

The final belief state is {S4, S8}. State S8 has both squares clean, but state S4 has a dirty left square. Therefore, the plan does not guarantee that the goal is reached in every possible state.